

08 — ACID Properties & Isolation

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1. COMMIT — Deposit

The screenshot displays a database IDE window titled "08 - ACID Properties & Isolation". It shows a sequence of SQL queries and their execution results.

Query 1:

```
7 WHERE Account_No = 10001;
8
9 • SELECT *
10 FROM Account
11 WHERE Account_No = 10001;
12
13 • COMMIT;
14
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad

Account 1

Output:

#	Time	Action	Message
1	12:36:49	USE BankDB	0 row(s) affected
2	12:36:50	START TRANSACTION	0 row(s) affected
3	12:36:52	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 C
4	12:36:54	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

Query 2:

```
10 FROM Account
11 WHERE Account_No = 10001;
12
13 • COMMIT;
14
15 • SELECT *
16 FROM Account
17 WHERE Account_No = 10001;
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad

Account 2

Output:

#	Time	Action	Message
1	12:36:49	USE BankDB	0 row(s) affected
2	12:36:50	START TRANSACTION	0 row(s) affected
3	12:36:52	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows
4	12:36:54	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
5	12:37:27	COMMIT	0 row(s) affected
6	12:37:35	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

2. ROLLBACK — Cancel Withdrawal

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Limit to 1000 rows

```
20
21 • UPDATE Account
22   SET Balance = Balance - 10000
23   WHERE Account_No = 10001;
24
25 • SELECT *
26   FROM Account
27   WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: IA

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 3 x

Output

Action Output

#	Time	Action	Message
✓ 4	12:36:54	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 5	12:37:27	COMMIT	0 row(s) affected
✓ 6	12:37:35	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 7	13:28:51	START TRANSACTION	0 row(s) affected
✓ 8	13:28:54	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched
✓ 9	13:28:59	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

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Limit to 1000 rows

```
26   FROM Account
27   WHERE Account_No = 10001;
28
29 • ROLLBACK;
30
31 • SELECT *
32   FROM Account
33   WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: IA

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 4 x

Output

Action Output

#	Time	Action	Message
✓ 6	12:37:35	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 7	13:28:51	START TRANSACTION	0 row(s) affected
✓ 8	13:28:54	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected R
✓ 9	13:28:59	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 10	13:29:15	ROLLBACK	0 row(s) affected
✓ 11	13:29:18	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

3. SAVEPOINT — Partial Rollback

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```
50 SET Balance = Balance + 10000
51 WHERE Account_No = 10003;
52
53 ROLLBACK TO SAVEPOINT Withdrawal1;
54
55 COMMIT;
56
57 SELECT *
58 FROM Account;
```

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	65000.00	Hyderabad
	10002	102	Savings	92000.00	Hyderabad
	10003	103	Current	100000.00	Bangalore
	10004	104	Savings	50000.00	Chennai
	10005	105	Current	80000.00	Hyderabad
	10006	106	Savings	77000.00	Delhi
	10007	107	Current	175000.00	Mumbai
	10008	108	Savings	30000.00	Pune
	10009	109	Savings	105000.00	Hyderabad
	10010	110	Current	95000.00	Vijayawada
*	NULL	NULL	NULL	NULL	NULL

Account 5

Output

Action Output

#	Time	Action	Message
✓ 15	13:31:24	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Change
✓ 16	13:31:26	SAVEPOINT Withdrawal1	0 row(s) affected
✓ 17	13:31:29	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Change
✓ 18	13:31:32	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
✓ 19	13:31:35	COMMIT	0 row(s) affected
✓ 20	13:31:37	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned

4. BANK TRANSFER — COMMIT

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```
66 UPDATE Account
67 SET Balance = Balance + 15000
68 WHERE Account_No = 10002;
69
70 SELECT *
71 FROM Account
72 WHERE Account_No IN (10001, 10002);
73
74 COMMIT;
```

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	107000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 6

Output

Action Output

#	Time	Action	Message
✓ 19	13:31:35	COMMIT	0 row(s) affected
✓ 20	13:31:37	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
✓ 21	13:32:30	START TRANSACTION	0 row(s) affected
✓ 22	13:32:32	UPDATE Account SET Balance = Balance - 15000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Change
✓ 23	13:32:34	UPDATE Account SET Balance = Balance + 15000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Change
✓ 24	13:32:36	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned

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```

72 WHERE Account_No IN (10001, 10002);
73
74 • COMMIT;
75
76 • SELECT
77     Account_No,
78     Balance
79 FROM Account
80 WHERE Account_No IN (10001, 10002);

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: 1A

	Account_No	Balance
▶	10001	50000.00
	10002	107000.00
•	NULL	NULL

5. CHECK CURRENT ISOLATION LEVEL

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```

75
76 • SELECT
77     Account_No,
78     Balance
79 FROM Account
80 WHERE Account_No IN (10001, 10002);
81
82 • SELECT @@SESSION.transaction_isolation;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: 1A

	@@SESSION.transaction_isolation
▶	REPEATABLE-READ

Result 8

Output

Action Output

#	Time	Action	Message
✓ 22	13:32:32	UPDATE Account SET Balance = Balance - 15000 WHERE Account_No = 10001	1 row(s) affected R
✓ 23	13:32:34	UPDATE Account SET Balance = Balance + 15000 WHERE Account_No = 10002	1 row(s) affected R
✓ 24	13:32:36	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned
✓ 25	13:33:03	COMMIT	0 row(s) affected
✓ 26	13:33:06	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned
✓ 27	13:34:51	SELECT @@SESSION.transaction_isolation LIMIT 0, 1000	1 row(s) returned